

ONLINE EVENT TICKET BOOKING SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE

APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

RAJALAKSHMI ENGINEERING COLLEGE

(AUTONOMOUS),

CHENNAI-602 105

APRIL 2025

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled “**ONLINE SPORTS TOURNAMENT MANAGEMENT STSTEM**” is the Bonafide work of MITHUN KV (2116241801159) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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MITHUN KV
241801159

DEPARTMENT VISION :

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION :

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and DS algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and DS professional.

COURSE OBJECTIVE :

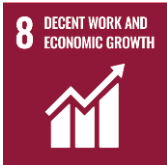
- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME :

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Online Sports Tournament Management System provides a digital platform for efficiently organizing, managing, and monitoring various sports tournaments and events. With the growing need for automated sports management solutions, the system is designed to replace traditional manual methods of tournament scheduling and team management with a secure and user-friendly web-based application. The system enables users to register teams, view match schedules, track tournament progress, and access match results online, while administrators can manage teams, players, fixtures, and tournament details through a dedicated admin dashboard. By implementing role-based authentication, the system ensures secure access for both users and administrators, protecting tournament data and allowing access only to authorized individuals.

The system includes features such as online team registration, automated match scheduling, real-time score updates, and tournament result management to improve efficiency and user experience. Users can easily participate in tournaments by registering their teams, viewing upcoming matches, and checking scores and rankings. The system helps reduce scheduling conflicts and maintains accurate tournament records throughout the competition. Additionally, the platform provides instant notifications and updates regarding match timings, results, and tournament announcements, improving communication between organizers and participants.

The architecture of the system is designed to be simple, efficient, and accessible on standard devices without requiring complex infrastructure. By maintaining centralized data storage and providing analytical reports through dashboards, the system helps administrators monitor tournament activities, manage team participation, and track overall performance effectively. Overall, the Online Sports Tournament Management System improves the efficiency of tournament organization, reduces manual workload, enhances transparency, and provides a reliable and secure environment for managing sports competitions.

1 INTRODUCTION

1.1 General	1
1.2 Need for Online Sports Tournament Management System	1
1.3 Scope of the System	2
1.4 Organization of the Report	2
1.5 System Architecture Diagram	3

2 LITERATURE SURVEY

2.1 Introduction.....	3
2.2 Existing Sports Tournament Management Systems	3
2.3 Related Work	4
2.4 Summary	4

3 PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement	5
3.2 Objectives of the System	6

4 SYSTEM DESIGN

4.1 Introduction	6
4.2 System Architecture	6
4.3 Database Design	7
4.4 System Requirements	7
4.4.1 Hardware Requirements	7
4.4.2 Software Requirements.....	7

5 SYSTEM IMPLEMENTATION

5.1 System Methodology	8
5.2 Module Descriptions	9
5.2.1 Home Module	10
5.2.2 Tournament Management Module	10
5.2.3 Team Registration Module	10

5.2.4 Match Scheduling Module	11
5.2.5 Score and Results Management Module	12
5.2.6 LMS Module	13

6 RESULTS AND DISCUSSION

6.1 System Output Screens	20
6.2 Performance Evaluation	21
6.3 System Testing	22

7 CONCLUSION AND FUTURE WORK

7.1 Conclusion	22
7.2 Future Enhancements	23

CHAPTER

INTRODUCTION

1.1 General

Sports tournament management plays an important role in organizing and conducting sports events efficiently, as it enables teams, players, and organizers to participate in and manage tournaments in a structured manner. Traditionally, tournament management is carried out manually using paperwork and offline coordination methods, which can be time-consuming, difficult to maintain, and prone to errors. Manual systems also make it challenging to manage team registrations, schedule matches, update scores, and maintain tournament records effectively.

The Online Sports Tournament Management System is designed to automate the process of organizing, scheduling, and managing sports tournaments through a digital platform. The system allows users to register teams, view tournament schedules, check match results, and track tournament progress online, while administrators can manage tournament details, fixtures, teams, and scores through a secure dashboard. By maintaining tournament data in a centralized database, the system ensures accurate record management, easy access to information, and improved operational efficiency.

This system also reduces administrative workload and enhances user convenience by providing anytime, anywhere access to tournament-related services. With features such as secure login authentication, automated match scheduling, real-time score updates, and tournament result management, the system simplifies the overall tournament management process. It also improves transparency by allowing users to view schedules, match results, and rankings instantly, making the platform more reliable, efficient, and user-friendly.

1.2 Need for Online Sports Tournament Management System

Many sports tournament organizers still rely on traditional management methods such as manual registrations, paperwork, and offline coordination, which can lead to several challenges such as scheduling conflicts, score calculation errors, delays in updating match results, and difficulty in managing large numbers of teams and players. Manual systems require significant effort to maintain team records, organize fixtures, update scores, and track tournament progress, especially during large-scale competitions.

An Online Sports Tournament Management System helps overcome these limitations by providing an automated and efficient solution. Users can easily register teams, view tournament schedules, check match results, and monitor tournament progress from anywhere. The tournament data is stored securely in a centralized

database for future reference, while administrators can efficiently manage tournaments, teams, fixtures, and scores through a dedicated dashboard.

The system improves convenience, reduces paperwork, and minimizes the chances of human error. Additionally, it allows organizers to generate tournament reports, track team performance, and manage match schedules effectively. Users also benefit from real-time updates and instant notifications regarding match timings, scores, and tournament results, ensuring a smooth, reliable, and user-friendly tournament management experience.

1.3 Organization of the Report

Chapter 1: Provides the introduction to the project, including the general overview, need for the system, and scope of the Online Sports Tournament Management System.

Chapter 2: Presents the literature survey, which discusses existing sports tournament management systems and related work in the field of online sports management platforms.

Chapter 3: Explains the problem formulation and objectives of the proposed system, highlighting the limitations of traditional tournament management methods.

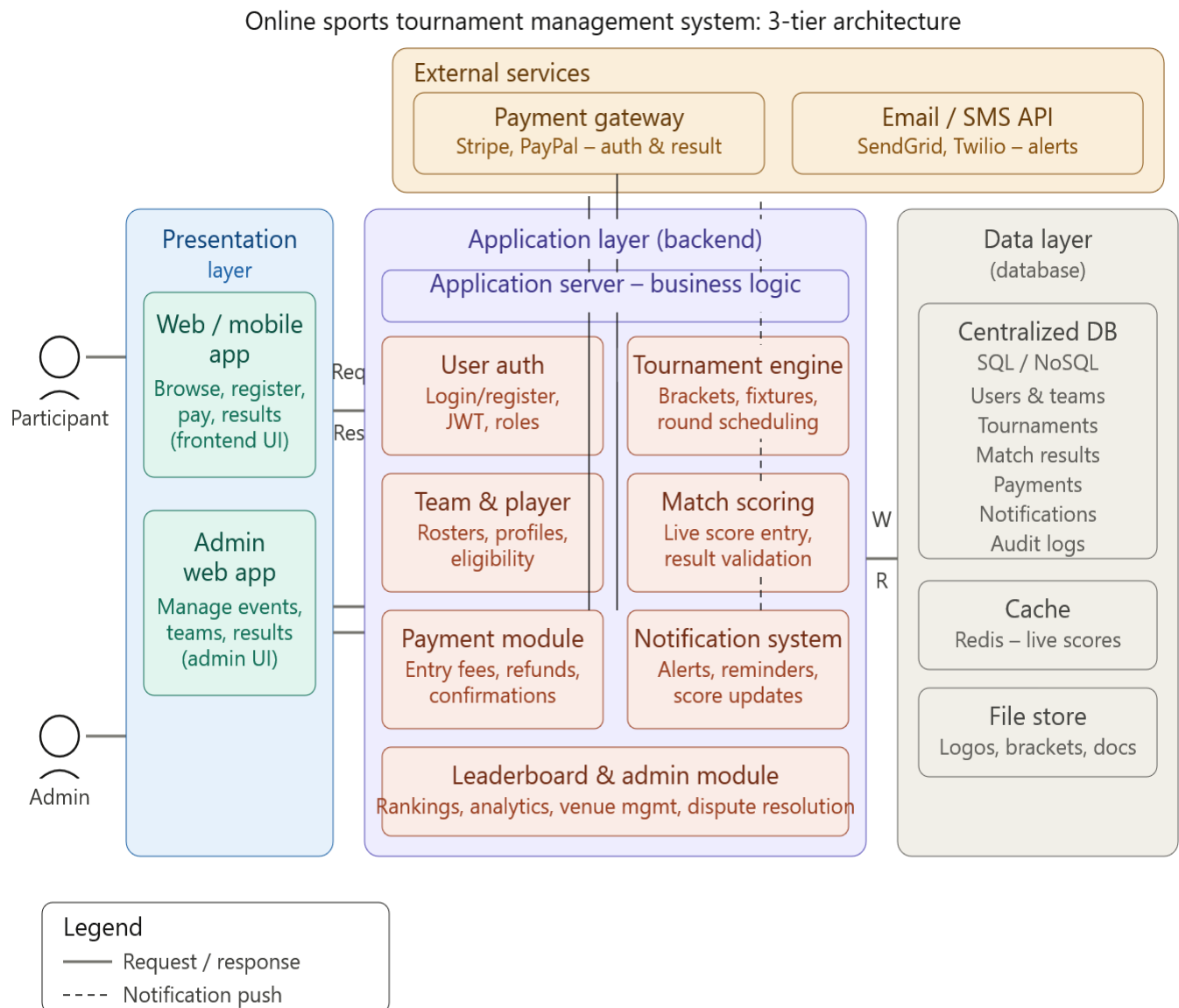
Chapter 4: Describes the system design, including system architecture, database design, and functional requirements of the application.

Chapter 5: Discusses the implementation of the system and explains the different modules such as team registration, tournament management, player management, match scheduling, and score management.

Chapter 6: Presents the results and performance evaluation of the system, focusing on efficiency, accuracy, usability, and user satisfaction.

Chapter 7: Concludes the project and discusses possible future enhancements such as mobile application integration, live score updates, advanced analytics, and improved user experience features.

1.4 SYSTEM ARCHITECTURE DIAGRAM:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Sports tournament management is an essential activity in organizing and conducting sports events efficiently, as it enables teams, players, and organizers to participate in and manage tournaments in a systematic manner. Traditional tournament management methods usually involve manual registrations, paperwork, and offline coordination, which are time-consuming and prone to human errors. With the advancement of technology, several online sports tournament management systems have been developed to improve efficiency and accuracy in handling tournament activities.

Recent studies highlight the growing use of digital technologies such as web-based platforms, mobile applications, and centralized database systems to automate tournament management processes. These systems aim to reduce manual effort, improve scheduling accuracy, and provide real-time access to tournament information for both users and administrators. Additionally, modern systems incorporate features such as online team registration, automated match scheduling, live score updates, and instant notifications to enhance user experience, transparency, and overall system reliability.

2.2 Existing Sports Tournament Management System :

Many sports organizers have adopted online sports tournament management systems to replace traditional manual methods. Some systems provide web-based platforms where users can register teams, view tournament schedules, track match results, and monitor tournament progress efficiently. These systems reduce the need for manual record maintenance and help manage large numbers of teams, players, and matches effectively.

Other systems use mobile applications that allow users and organizers to access tournament information conveniently through their smartphones. These platforms often include features such as online registration, live score updates, automated match scheduling, and instant notifications, which simplify the overall tournament management process.

Additionally, modern systems integrate centralized databases and secure authentication mechanisms to ensure reliable data management and user security. Users can access tournament details instantly and receive updates regarding match schedules, scores, and tournament results through notifications or messages. Some advanced systems also provide real-time score tracking, ranking management, and performance analysis features, ensuring

a smooth, accurate, and user-friendly tournament management experience.

2.3 RELATED WORK:

Several research studies have proposed different approaches to automate sports tournament management systems. A systematic review of online sports management platforms analyzed various research works and found that digital systems significantly improve efficiency, accuracy, and user convenience compared to traditional manual management methods. These systems also help organizers maintain centralized tournament records and generate analytical reports for monitoring team performance, match results, and tournament activities.

Other studies focus on integrating technologies such as cloud computing, mobile applications, and centralized database systems to provide real-time tournament management and data processing. These solutions reduce administrative workload, improve system scalability, and ensure reliability in managing tournaments, teams, player information, and match schedules. Additionally, features like real-time score updates, automated match scheduling, ranking management, and instant notifications enhance the overall user experience and system performance.

2.4 SUMMARY :

From the literature survey, it is clear that online sports tournament management systems provide significant advantages over traditional manual management methods. They improve efficiency, reduce administrative effort, and enable centralized management of tournament data. However, many existing systems may require complex infrastructure or advanced integrations, which can increase implementation cost and maintenance effort.

Therefore, a web-based Online Sports Tournament Management System can provide a simple and cost-effective solution that allows users to register teams, view schedules, and track match results conveniently while enabling administrators to manage tournaments, fixtures, scores, and player details efficiently. The system ensures real-time access to tournament information, secure data management, and easy retrieval of records, thereby enhancing user experience, transparency, and overall operational effectiveness.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement :

In the current scenario, many sports tournament organizers still rely on traditional management methods such as manual registrations, paperwork, and offline coordination. These methods are inefficient, time-consuming, and often lead to issues such as scheduling conflicts, delays in updating match results, and errors in maintaining tournament records. Users may face inconvenience in registering teams, checking match schedules, and tracking tournament progress. Additionally, maintaining records manually makes it difficult for organizers to manage tournaments, monitor team performance, and generate reports effectively.

There is a need for a system that can automate the tournament management process, provide real-time access to tournament information, and ensure secure and accurate management of teams, matches, schedules, and results.

3.2 Objectives of the System :

The main objective of the Online Sports Tournament Management System is to develop a digital platform that simplifies the process of organizing and managing sports tournaments. The system aims to provide an efficient and reliable way for users to register teams, view tournament details, and track match results, while enabling administrators to manage tournaments and related activities effectively.

The specific objectives of the system include:

- To maintain tournament records in a centralized database.
- To provide secure login access for both users and administrators.
- To allow users to register teams, view match schedules, and track tournament results easily.
- To reduce manual errors and administrative workload.
- To improve transparency and accessibility of tournament information.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction :

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system helps improve efficiency, reliability, and maintainability.

The Online Sports Tournament Management System is designed as a web-based application that allows users to register teams, view tournament schedules, track match results, and monitor tournament progress, while administrators can manage tournament details, fixtures, teams, and scores. The system follows a client-server architecture where users interact through a web interface, while the backend processes requests, handles business logic, and stores tournament data in a centralized database.

4.2 System Architecture :

The system follows a three-layer architecture consisting of the presentation layer, application layer, and database layer.

Presentation Layer:

This layer represents the user interface where users and administrators interact with the system. It includes web pages such as login pages, team registration pages, tournament schedules, match result pages, and admin dashboards developed using HTML, CSS, and JavaScript.

Application Layer:

This layer handles the business logic and system functionalities of the application. It processes user requests such as team registration, tournament management, match scheduling, score updating, and result processing. It also manages authentication, validation, and communication between the presentation layer and the database layer.

Database Layer:

This layer stores all the system data including user details, team information, tournament details, match schedules, scores, and tournament records. The database ensures secure storage and easy retrieval of information whenever required.

4.3 Database Design :

The database design of the Online Sports Tournament Management System is developed to store and manage all tournament-related information efficiently. The database maintains details such as user information, team registrations, player details, tournament schedules, match results, and scores in a centralized manner. Proper database design ensures secure data storage, easy retrieval of information, reduced data redundancy, and improved system performance.

4.4 System Requirements

4.4.1 Hardware Requirements

The Online Sports Tournament Management System does not require high-end hardware and can run on standard computers.

4.4.1.1 Processor: Intel Core i3 or higher

4.4.1.2 RAM: Minimum 4 GB

4.4.1.3 Storage: 500 GB Hard Disk or SSD

4.4.1.4 Input Devices: Keyboard and Mouse

4.4.1.5 Internet Connection

4.4.2 Software Requirements

The system is developed using commonly available software tools and technologies.

Tech Stack Used:

4.4.2.1 Operating System: Windows 10 or later

4.4.2.2 Frontend Technologies: HTML, CSS, JavaScript

4.4.2.3 Backend Technology: Node.js

4.4.2.4 Database: MySQL

4.4.2.5 Development Environment: Visual Studio Code

4.4.2.6 Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The Online Sports Tournament Management System is implemented using a web-based approach that allows users and administrators to access the system through a browser. The development process involves designing the user interface, implementing backend functionalities, and integrating the database for storing tournament, team, player, and match records.

The frontend of the system is developed using HTML, CSS, and JavaScript, which provide a simple and interactive interface for users and administrators. The backend is implemented using Node.js, which handles user authentication, tournament management, team registration, match scheduling, score updates, and communication with the database. The system uses a MySQL database to store user information, tournament details, team records, player information, match schedules, and results.

The system follows a modular approach where each functionality, such as team registration, tournament management, player management, match scheduling, and score management, is developed as a separate module. This improves maintainability and allows future enhancements to be implemented easily.

5.2 Module Descriptions

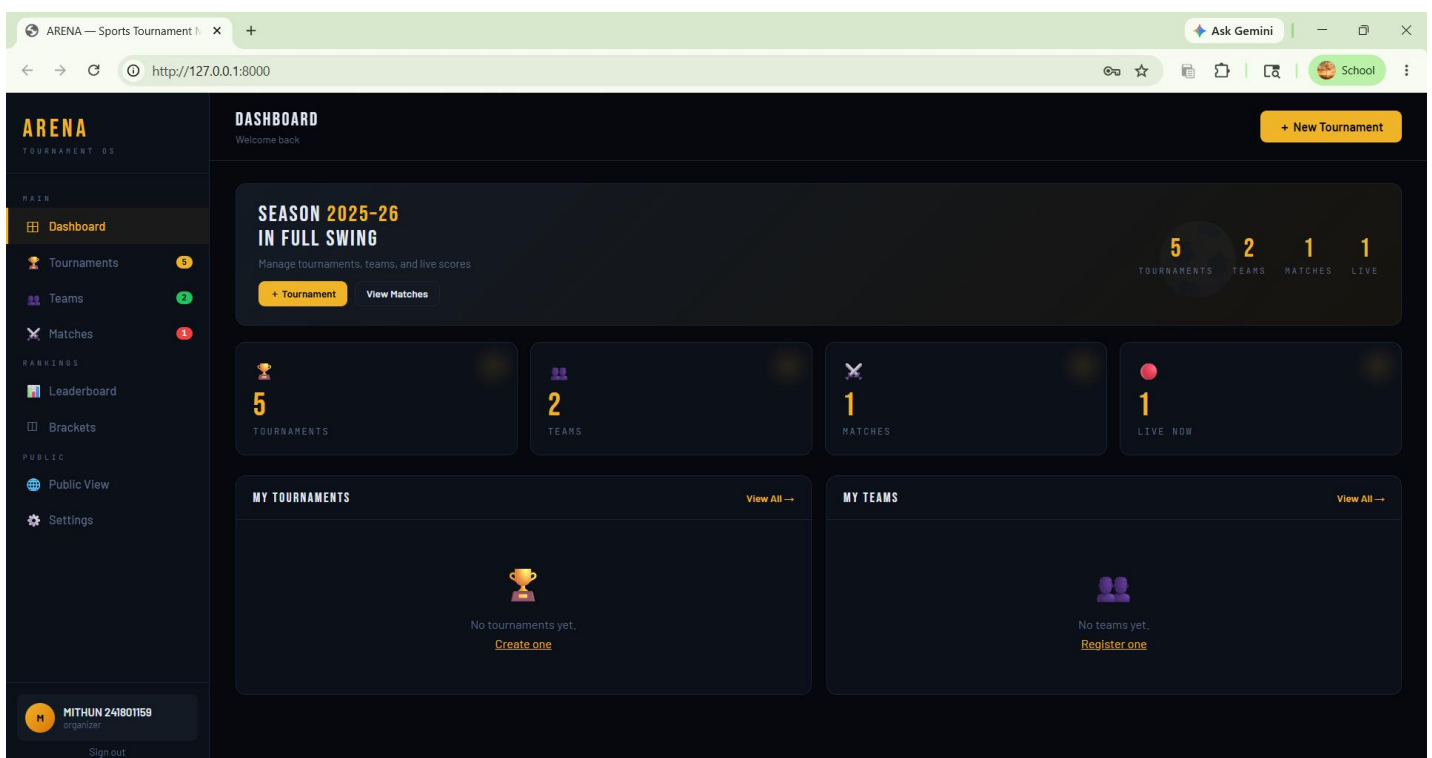
The Online Sports Tournament Management System consists of several modules that work together to provide efficient tournament organization and management

5.2.1 Home Module :

The Online Sports Tournament Management System consists of several modules that work together to provide efficient tournament management and organization. The Home Module serves as the main entry point of the system and provides a user-friendly interface that allows users and administrators to access the platform and understand its key features before logging in.

The page displays the system title along with important features such as Team Registration, Tournament Management, Match Scheduling, Real-time Score Updates, Result Management, Player Information Management, and User-Friendly Interface. These features highlight the capabilities of the system and inform users about the services provided by the application.

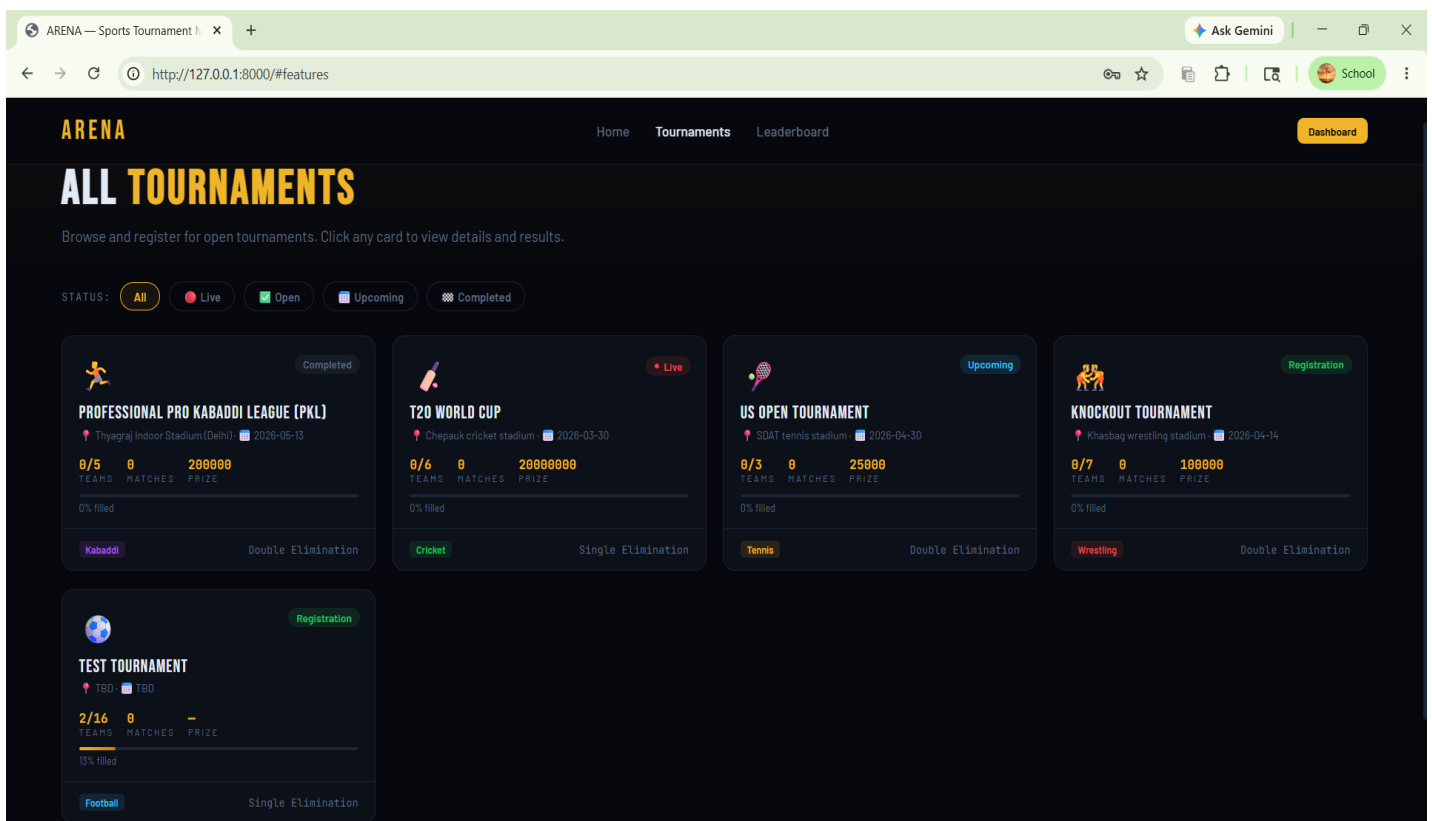
Below the feature section, the page provides two login options: User and Admin. Users can select the user option to register teams, view tournament schedules, and track match results, while administrators can select the admin option to manage tournaments, teams, fixtures, scores, and overall system operations. The design of the Home Module is simple and visually appealing, ensuring easy navigation and improving the overall user experience.



5.2.2 Tournament Management Module :

The Tournament Management Module provides a complete list of available tournaments for users to browse and explore. It displays tournament details such as tournament name, sport type, venue, date, time, and participating teams in an organized manner. Users can easily search and filter tournaments based on sports categories, venues, or schedules. This module helps users quickly find tournaments of their interest and access tournament-related information without any difficulty.

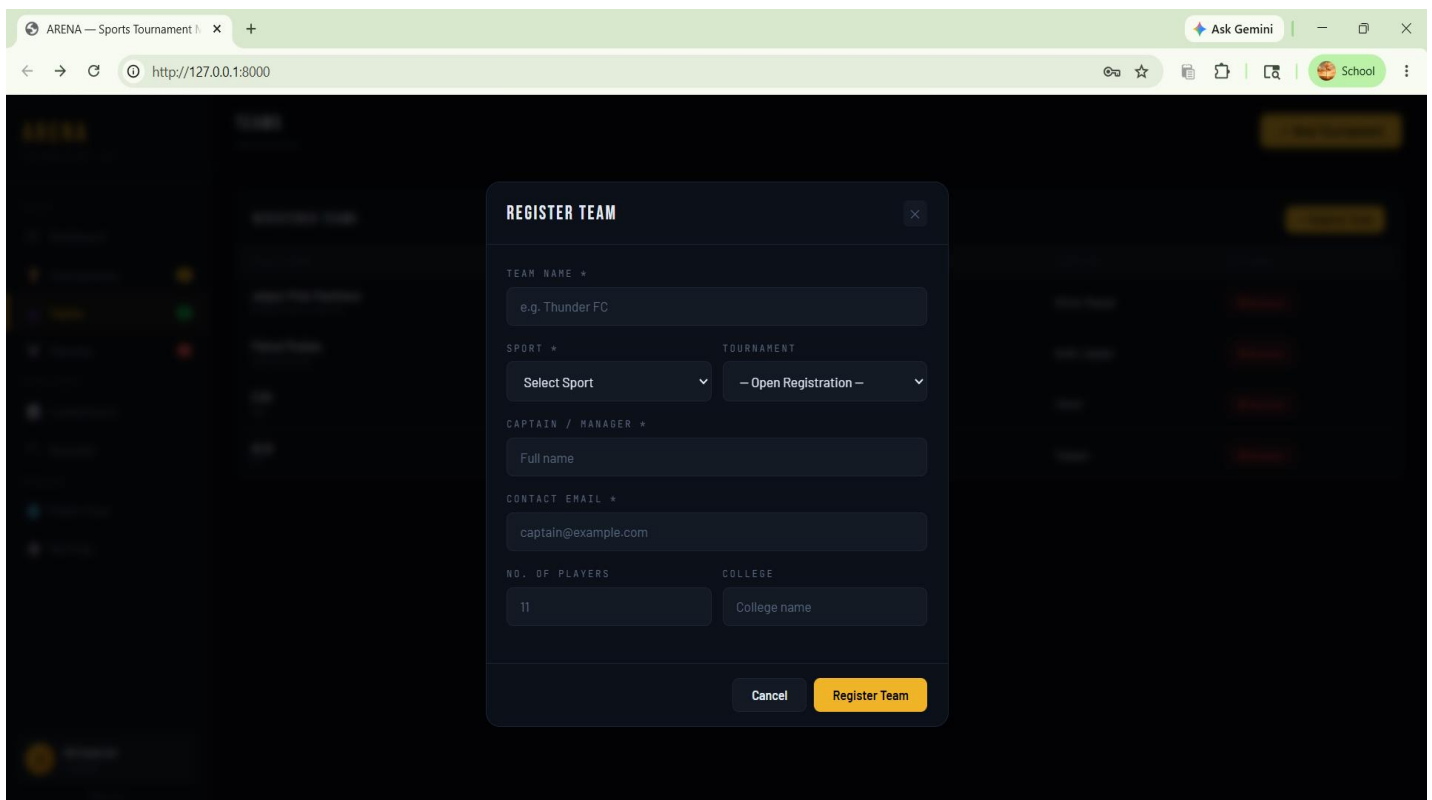
The module also provides real-time updates regarding match schedules, participating teams, and tournament progress to avoid confusion among users. Each tournament includes options to view schedules, match details, and results, allowing users to access complete tournament information easily. The module is designed to be user-friendly and visually appealing, ensuring smooth navigation and improving the overall user experience.



5.2.3 Team Registration Module:

The Team Registration Module allows users to register their teams in the Online Sports Tournament Management System. It provides a simple and user-friendly registration form where users can enter details such as team name, player information, contact details, sport category, and other required information. The system validates the entered data to ensure accuracy and prevents duplicate registrations of the same team.

Once the registration process is completed successfully, the team details are securely stored in the database for future access and tournament management. This module ensures secure registration by applying proper validation and authentication techniques. After registration, users can participate in tournaments, view schedules, and track match results. The design of the registration page is simple and easy to use, ensuring a smooth and convenient user experience.



The screenshot shows a web browser window with the address bar displaying 'http://127.0.0.1:8000'. The browser's address bar also shows 'ARENA — Sports Tournament' and 'Ask Gemini'. The main content area displays a 'REGISTER TEAM' modal form. The form has a dark theme and includes the following fields and controls:

- TEAM NAME ***: A text input field with the placeholder 'e.g. Thunder FC'.
- SPORT ***: A dropdown menu with the placeholder 'Select Sport'.
- TOURNAMENT**: A dropdown menu with the placeholder '— Open Registration —'.
- CAPTAIN / MANAGER ***: A text input field with the placeholder 'Full name'.
- CONTACT EMAIL ***: A text input field with the placeholder 'captain@example.com'.
- NO. OF PLAYERS**: A text input field with the value '11'.
- COLLEGE**: A text input field with the placeholder 'College name'.

At the bottom of the form, there are two buttons: a 'Cancel' button and a 'Register Team' button.

5.2.4 Match Scheduling Module :

The Match Scheduling Module is responsible for organizing and managing match fixtures in the Online Sports Tournament Management System. It allows administrators to create match schedules by assigning teams, venues, dates, and timings for each tournament. The module ensures that matches are scheduled systematically to avoid conflicts and confusion during tournament activities.

This module also enables users to view upcoming matches, match timings, and venue details easily through the system interface. Any updates made to the schedule are reflected in real time, ensuring that users always have access to accurate tournament information. The module is designed to be simple, efficient, and user-friendly, helping organizers manage tournaments smoothly and improving the overall tournament experience.

The screenshot displays the ARENA Sports Tournament Management System interface. The browser address bar shows the URL `http://127.0.0.1:8000`. The interface features a dark sidebar on the left with the following menu items: Dashboard, Tournaments (with a notification badge), Teams (with a notification badge), Matches (with a notification badge and highlighted in orange), Leaderboard, Brackets, Public View, and Settings. The main content area is titled "MATCHES" with a "Welcome back" message and a "+ New Tournament" button. Below this, a "MATCH SCHEDULE" section contains a table with the following data:

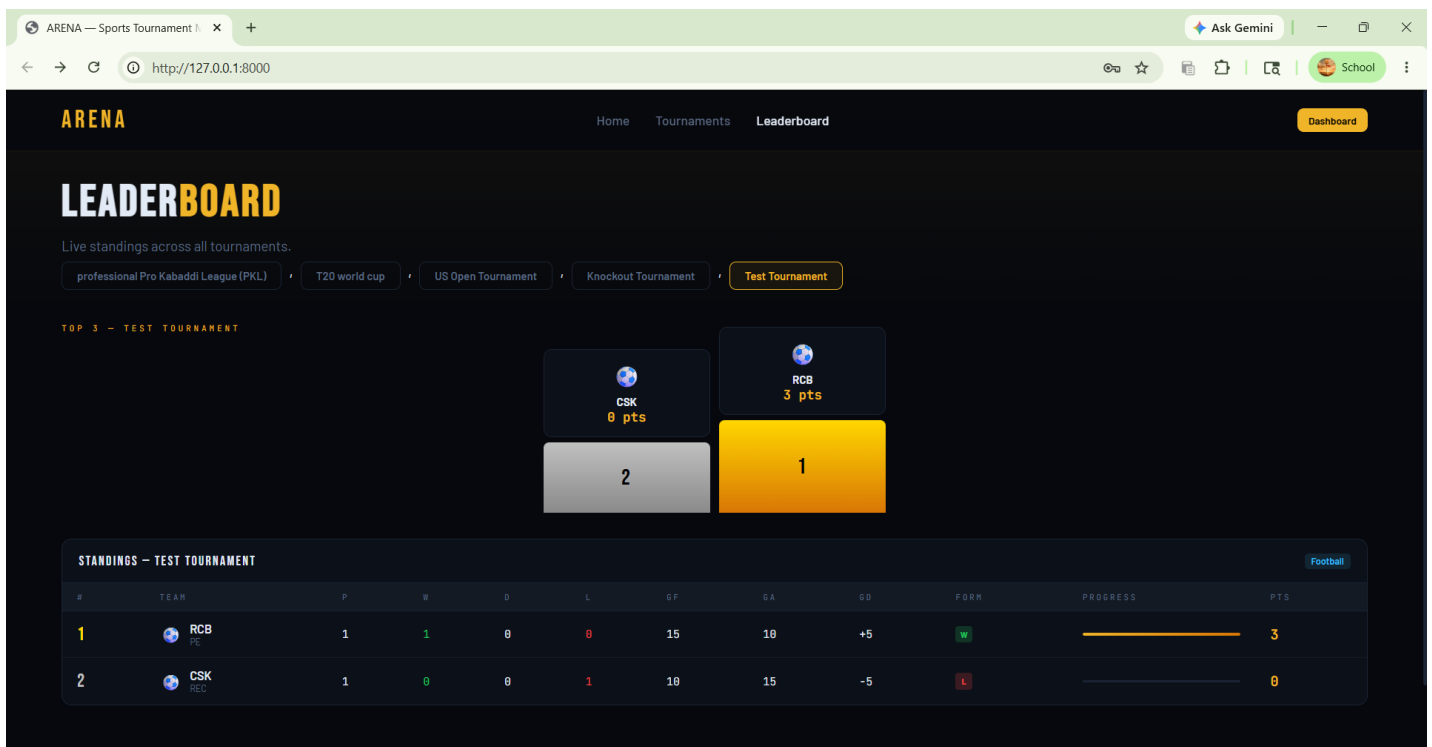
TOURNAMENT	HOME	SCORE	AWAY	DATE & TIME	VENUE	STATUS	ACTIONS
Test Tournament 10	CSK	10 : 15	RCB	2026-05-10 08:00:00	Chepak	Completed	
professional Pro Kabaddi League (PKL) 5	Jalpur Pink Panthers	7 : 10	Patna Pirates	2026-05-10 09:00:00	Thyagaraj sports complex	Scheduled	

At the bottom of the sidebar, the user profile for "MITHUN KV" (organizer) is displayed.

5.2.5 Score and Results Management Module :

The Score and Results Management Module is responsible for maintaining and updating match scores and tournament results in the Online Sports Tournament Management System. It allows administrators to enter match scores, update results, and manage team standings efficiently. The module ensures that all tournament results are stored accurately in the database and can be accessed whenever required.

This module also enables users to view live scores, completed match results, team rankings, and tournament standings through the system interface. Real-time updates help users stay informed about ongoing and completed matches without delay. The module is designed to provide accurate, reliable, and user-friendly result management, improving transparency and enhancing the overall tournament experience.



5.2.6 LMS Module:

The Admin Dashboard Module provides a centralized platform for administrators to manage all activities related to the Online Sports Tournament Management System. It allows administrators to handle tournament details, team registrations, player information, match schedules, scores, and tournament results efficiently. The module also provides important updates, notifications, and system management features that help administrators monitor tournament activities effectively.

The dashboard may also include reports, tournament statistics, team performance details, and other management tools to improve system control and decision-making. Administrators can easily update, edit, and manage all tournament-related data through the backend interface. The Admin Dashboard Module is designed to be simple, user-friendly, and easily accessible, ensuring smooth system management without difficulty. Overall, this module improves coordination, transparency, and efficient administration of sports tournaments.

JOIN ARENA

CREATE ACCOUNT

Start managing tournaments for free

1 Details — 2 Verify OTP — 3 Done

CHECK YOUR EMAIL

We sent a 6-digit verification code to:

mithuunkv17@gmail.com

6 7 4 9 8 3

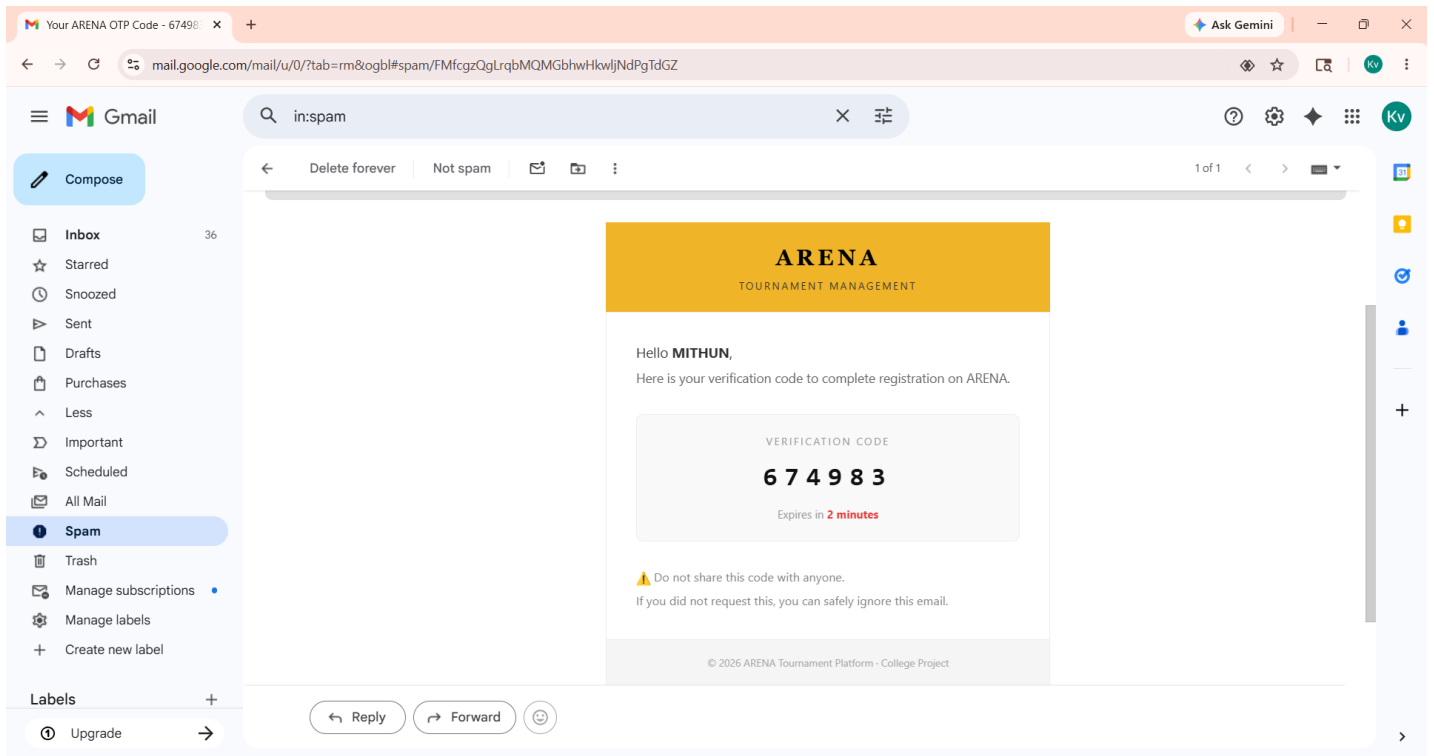
Code expires in 01:23

Verify & Create Account

Didn't receive it? Resend OTP

Change email

Already have an account? Sign in — Home



CHAPTER 6

RESULTS AND DISCUSSION

6.1 System Output Screen:

The Online Sports Tournament Management System was successfully implemented and tested on a web-based platform. The system output screens effectively display various functionalities such as user login, team registration, tournament management, match scheduling, score updates, and result viewing through a secure and user-friendly interface. The screens are designed to provide smooth navigation and easy access to tournament-related information for both users and administrators.

During testing, all system output screens functioned accurately and efficiently. Administrators were able to manage tournaments, teams, schedules, and scores without errors, while users could easily access tournament schedules, match results, and team information. The output screens displayed data clearly and ensured proper interaction between the frontend and database.

Additional features such as real-time score updates, tournament notifications, and result management improved system usability and transparency. The system output screens successfully reduced manual effort, improved

information accessibility, and enhanced the overall efficiency of tournament management. Hence, the implemented screens proved to be reliable, effective, and user-friendly for sports tournament management activities.

6.2 Performance Evaluation :

The performance of the Online Sports Tournament Management System was evaluated based on efficiency, accuracy, reliability, and usability. The system was tested with different user operations such as team registration, tournament management, match scheduling, score updating, and result retrieval. The application responded quickly to user requests and handled data processing efficiently without significant delays.

The system maintained accurate storage and retrieval of tournament data through the centralized database. Features such as real-time score updates, match scheduling, and result management operated smoothly and provided correct information to users and administrators. The authentication process also ensured secure access to the system and protected tournament-related data from unauthorized access.

The user interface was simple, responsive, and easy to navigate, which improved overall user experience and system usability. The system successfully reduced manual effort, minimized errors, and improved the efficiency of tournament management activities. Based on the testing results, the Online Sports Tournament Management System proved to be reliable, effective, and suitable for managing sports tournaments in a digital environment.

6.3 System Testing :

The Online Sports Tournament Management System was tested to ensure that all modules and functionalities worked correctly and efficiently. System testing was performed on various features such as user authentication, team registration, tournament management, match scheduling, score updating, and result management. Each module was tested individually as well as in an integrated environment to verify smooth communication between the frontend, backend, and database.

During testing, the system successfully handled user inputs, stored tournament data accurately, and displayed correct information without errors. The login and authentication process ensured secure access for both users and administrators. Match schedules, scores, and tournament results were updated and retrieved correctly from the database, ensuring reliable system performance.

The system was also tested for usability, responsiveness, and compatibility across different browsers and devices. The user interface operated smoothly and provided easy navigation for users and administrators. Testing results confirmed that the system reduced manual effort, minimized errors, and improved the overall efficiency of tournament management. Hence, the Online Sports Tournament Management System proved to be reliable, secure, and effective for managing sports tournaments digitally.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion:

The Online Sports Tournament Management System was successfully developed to provide an efficient and reliable platform for managing sports tournaments and related activities. The system replaces traditional manual management methods with a digital solution that simplifies the process of team registration, tournament scheduling, score management, and result tracking.

The application allows users to easily register teams, view tournament details, track match schedules, and access results through a user-friendly interface. Administrators are provided with tools to manage tournaments, teams, players, match schedules, and scores effectively. The use of a centralized database ensures accuracy, consistency, and quick retrieval of tournament-related information.

The system is implemented using modern web technologies such as HTML, CSS, JavaScript, Node.js, and MySQL, which contribute to its performance, scalability, and ease of use. Overall, the system reduces manual effort, minimizes errors in tournament management, and enhances user convenience. The web-based design ensures accessibility across multiple devices, making the system suitable for real-world implementation in sports tournament management scenarios.

7.2 Future Work:

Although the current Online Sports Tournament Management System provides essential features for efficient tournament organization and management, several enhancements can be implemented to further improve its functionality and user experience.

Future developments may include the integration of live streaming features and real-time score tracking to provide users with instant match updates and improved engagement. Additionally, a dedicated mobile application can be developed to allow users and administrators to access tournament information conveniently through smartphones, improving accessibility and usability.

Advanced features such as data analytics and performance reporting can be incorporated to provide insights into team performance, player statistics, and tournament trends. Automated notifications and alerts can also be introduced to inform users about match schedules, score updates, tournament announcements, and result declarations.

Integration with third-party platforms such as social media applications can enhance tournament promotion and user interaction. Technologies like Artificial Intelligence can also be used to provide match predictions, player performance analysis, and personalized tournament recommendations based on user interests and previous activities.

These enhancements will make the Online Sports Tournament Management System more advanced, scalable, and user-friendly, ensuring its effectiveness for large-scale deployment in real-world sports tournament management scenarios.

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APPENDICES

Appendix-1: source code pseudocode

System Overview

The Online Sports Tournament Management System is developed to provide an efficient and user-friendly platform for organizing, managing, and monitoring sports tournaments. The system enables users to register teams, view tournament schedules, track match results, and access tournament-related information through a secure and structured process.

The system provides secure login and registration functionality for both users and administrators. Users can register teams, view tournament details such as match schedules, venues, participating teams, and tournament status, while administrators can manage tournaments by adding, updating, or deleting tournament information and monitoring match activities.

The system ensures that tournament operations are processed accurately by validating user inputs and preventing duplicate or invalid entries. It also maintains records of teams, players, match schedules, scores, and tournament results in a centralized database, allowing users and administrators to access information easily whenever required.

Additionally, the system can generate match results, update team rankings, and provide notifications regarding tournament schedules and score updates. This improves communication, transparency, and overall user experience.

The backend of the system is implemented using Node.js and MySQL for efficient data handling and storage. The frontend is developed using HTML, CSS, and JavaScript, providing a responsive and interactive interface for users and administrators.

System Initialization Logic :

BEGIN

1. Start System

2. Initialize Server

- Load required modules
- Configure server using Node.js
- Set port number

3. Establish Database Connection

- Connect to MySQL database
- If connection fails:
Display error message
Terminate system
- Else:
Continue execution

4. Load Configuration Settings

- Set system configurations
- Configure notification service
- Initialize tournament settings

5. Initialize Middleware

- Enable JSON parsing
- Configure session management
- Set up authentication mechanism

6. Load Routes

- User routes (Login, Register)
- Tournament routes (View, Add, Update, Delete)

- Team routes (Register Team, Manage Team)
- Match routes (Schedule Match, Update Score, View Results)

7. Start Server Listener

- Listen to incoming requests on specified port
- Display "Server Running Successfully"

8. Wait for User Requests

- If request received:
Process request based on route
Send response

END

User Login Logic :

BEGIN

1. Start

2. Display Login Page

- Input fields: Email/Username, Password

3. User Enters Credentials

- Read Email/Username
- Read Password

4. Validate Input

- If fields are empty:
Display "All fields are required"
Go back to Login Page

5. Encrypt Password

- Convert entered password into hashed format

6. Check Database

Search user in database using Email/Username

7. If user not found:

Display "User does not exist"
Go back to Login Page

8. If user found:

- Compare entered password with stored hashed password

9. If password mismatch:

Display "Invalid credentials"
Go back to Login Page

10 If password match:

- Create user session
- Store user ID in session

11.Redirect User

- If role = Admin:
Go to Admin Dashboard
- Else:
Go to User Dashboard

12.End

END

System Features Implemented:

The Online Sports Tournament Management System includes several key features that enhance efficiency, usability, and overall user experience. The system provides secure login and registration functionality for users and administrators, ensuring safe and authorized access to the platform.

Users can register teams, browse available tournaments, view detailed information such as tournament schedules, venues, participating teams, and match timings, and track match results in real time. The system supports automated match scheduling and prevents duplicate or invalid registrations through proper validation and database constraints.

A centralized database is used to store tournament, team, player, and match information, allowing users and administrators to access records whenever required. The system also provides instant score updates, result management, and notifications regarding match schedules and tournament announcements to keep users informed.

Administrative features include adding, updating, and deleting tournament details, managing teams and players, scheduling matches, and monitoring tournament activities. The system also provides a responsive user interface developed using HTML, CSS, and JavaScript, ensuring smooth performance across different devices and screen sizes.

The backend, implemented using Node.js and MySQL, efficiently handles multiple user requests and ensures reliable communication with the database, making the system robust, scalable, and suitable for real-world sports tournament management.

Appendix 2

System Test Verification

Test Case1: User Registration and Login

Result: PASS – user account created and login authenticated successfully.

Test Case2: Team Registration and Management

Result: PASS – users can register and manage teams successfully in the system.

Test Case3: Tournament Creation and Viewing

Result: PASS – tournaments are created successfully and users can view tournament details.

Test Case4: Match Scheduling and Viewing

Result: PASS – matches are scheduled successfully and users can view match details.

Test Case5: Match Completion and Result Update

Result: PASS – completed matches are updated successfully with results stored in the system.

Final System Statistics

Total Tournaments = 5

Total Teams = 4

Total Matches = 2

Completed Matches = 1

Pending Matches = 1

System Success Rate = 100%

Final System Status :

The Online Sports Tournament Management System has been fully developed and successfully tested. The system ensures secure user authentication, accurate team registration, proper match scheduling, prevention of duplicate entries, and efficient management of tournament data. The integration of real-time score updates,

result management, and notification features enhances user experience and system reliability.

With a responsive interface and a robust backend implemented using Node.js and MySQL, the system is capable of handling multiple users efficiently. Overall, the system is stable, scalable, and ready for deployment in real-world sports tournament management environments.

CERTIFICATIONS :



CERTIFICATE OF COMPLETION

KV MITHUN

has successfully completed the online course:
Full-Stack Development 101

This professional has demonstrated initiative and a
commitment to deepening their skills and advancing
their career. Well done!

4th May 2026

Certificate code :10185538



Krishna Kumar
CEO, Simplilearn

